Recommender Systems

News Recommendation with the MIND Dataset

Group 11 · TDT4215 · NTNU · 22.04.2026

Svein · Mehdi · Maryam · Frikk

Agenda

01	Introduction & Motivation
02	Theory: Content-Based, Collaborative & Hybrid
03	Exploratory Data Analysis (MIND Dataset)
04	Methods & Implementation
05	Evaluation Metrics
06	Results
07	Discussion & Conclusions

Introduction

- Recommender systems are central to modern digital platforms
- Dominant approaches: Content-Based Filtering (CBF) and Collaborative Filtering (CF)
- Goal: develop and compare multiple recommender strategies for news
- Benchmark: MIND (Microsoft News Dataset) – widely used in the field
- Research question: How do CBF, CF, and their hybrid compare on accuracy and beyond-accuracy metrics?

Dataset Snapshot

1M
Users

160k
Articles

2.2M
Train Sessions

4%
Overall CTR

4
Models Compared

Theory: Recommendation Approaches

Content-Based Filtering (CBF)

Recommends items similar to those a user liked.

Articles represented as TF-IDF vectors (title, abstract, category, subcategory).

User profile = recency-weighted sum of past reads.

✓ Handles cold-start well

Collaborative Filtering (CF)

Predicts preferences from user–item interaction patterns.

Builds sparse user-item matrix → Matrix factorization (SVD) → item embeddings.

Recency-weighted user profile from embeddings.

✗ Struggles with cold-start

Weighted Hybrid Filtering

Linear combination of CF and CBF scores:

$S_{final} = \omega_{cf} \times \mathrm{norm}(S_{cf}) + \omega_{cbf} \times \mathrm{norm}(S_{cbf})$

Scores normalized (min-max) before fusion.

Weights tuned on validation

Exploratory Data Analysis (MIND Small – Train)

Category distribution: news & sports dominate; remaining categories more evenly spread

Title lengths: consistently short (7–15 words); abstracts vary — many have no abstract

User sessions: impressions, clicks & history length all heavy-tailed (power-law)

Session timing: peak activity late morning, gradual decline through the day

Cold-start challenge: ~85% of articles received zero clicks in training

Key Stats

5.84 M
Impressions (train)

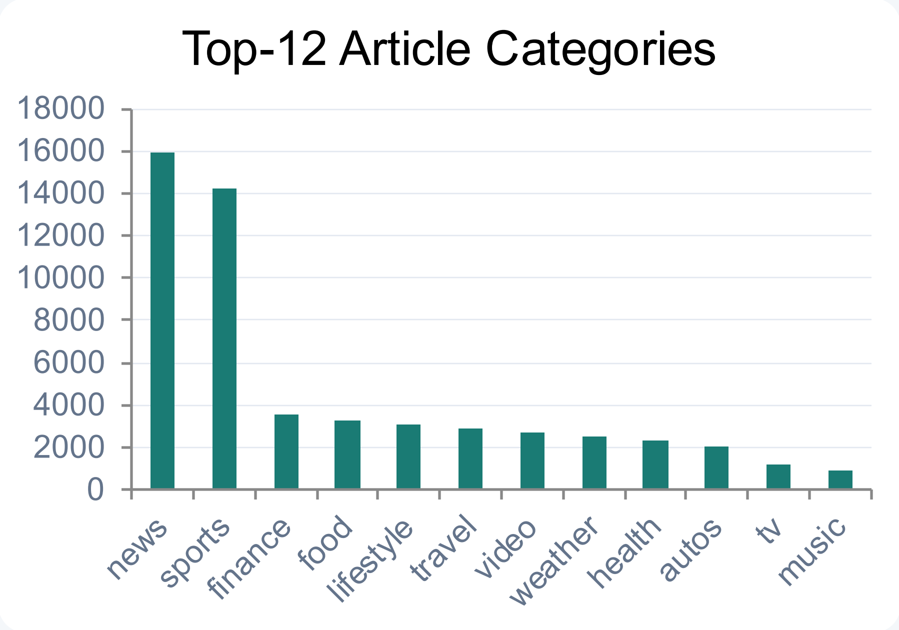
236 k
Clicks (train)

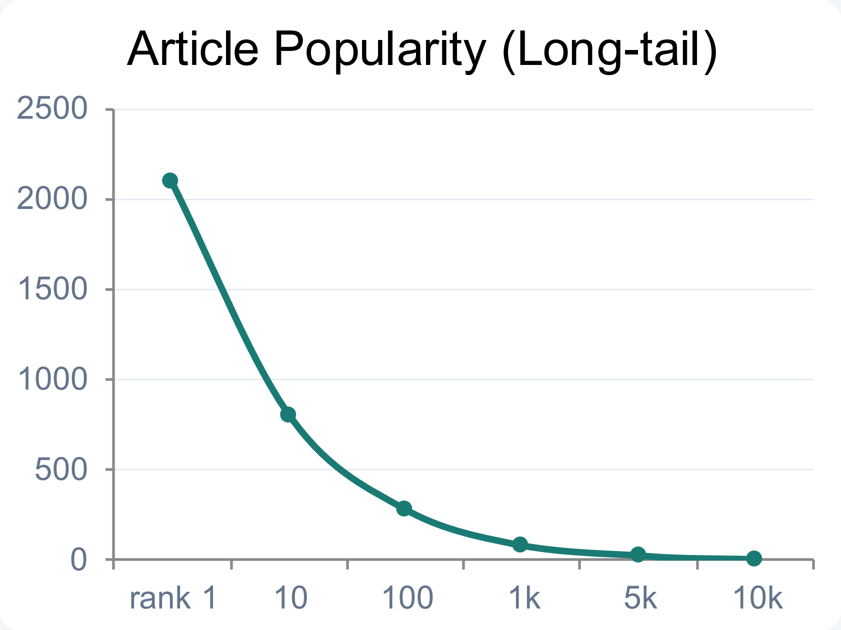
4.0%
Overall CTR

~32
Avg. history length

17
Article categories

EDA: Category Distribution & Article Popularity





Category imbalance and long-tail popularity distribution motivate beyond-accuracy metrics (novelty & diversity)

Methods & Implementation

Popularity Baseline	Counts training clicks per article. Non-personalized — identical scores for all users. Zero score for unseen articles.
Content-Based (CBF)	TF-IDF vectors (title + abstract + category + subcategory). Recency-weighted user profile. Cosine similarity scoring. Cold-start friendly.
Collaborative (CF)	Implicit user-item matrix → Truncated SVD (32 dims). Recency-weighted profile from item embeddings. Cannot score unseen items (cold-start limitation).
Hybrid	Weighted linear combination of CF & CBF scores. Min-max normalization before fusion. Weights tuned in 0.1 increments — best: CF = 0.1, CBF = 0.9.

Evaluation Metrics

Accuracy Metrics

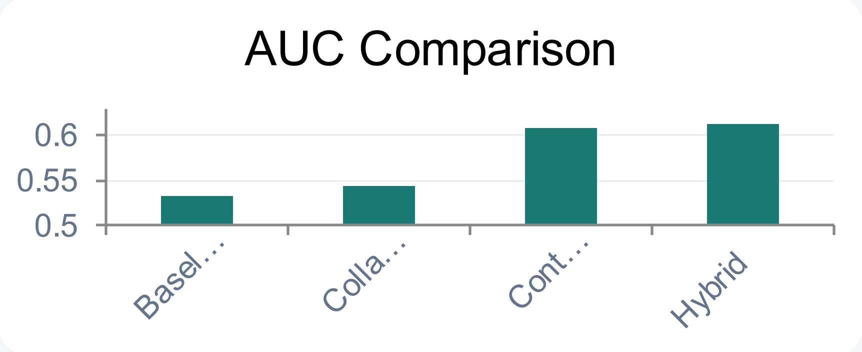
AUC	Overall ranking ability — clicked vs non-clicked
MRR	Mean Reciprocal Rank — rewards first relevant result
nDCG@5	Normalized discounted gain — top-5 position sensitivity
nDCG@10	Normalized discounted gain — top-10 position sensitivity

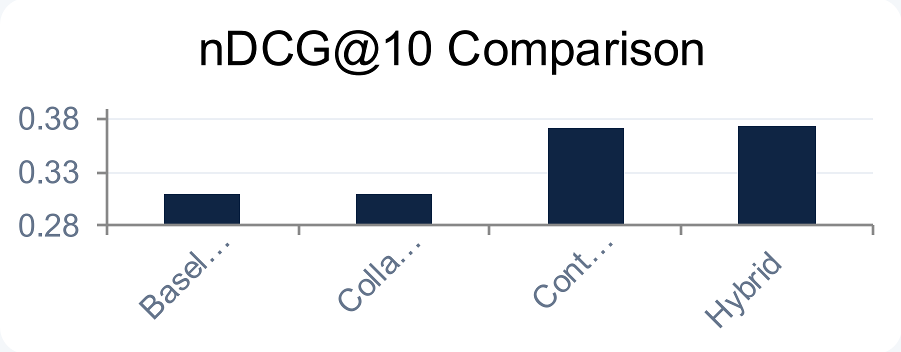
Beyond-Accuracy Metrics

Novelty	Mean self-information of top-10: $-\log_2 p(\text{item})$. Higher = less popular items recommended
Div/category	Fraction of distinct categories in top-10 recommendations
Div/subcategory	Fraction of distinct subcategories in top-10 recommendations

Results: MIND Small Development Set

Model	AUC	MRR	nDCG@5	nDCG@10	Div/cat	Div/sub	Novelty
Baseline	0.5318	0.2671	0.2460	0.3098	0.6898	0.8790	14.04
Collaborative	0.5429	0.2632	0.2473	0.3088	0.6774	0.8840	14.58
Content-Based	0.6073	0.3306	0.3108	0.3718	0.5810	0.7981	16.03
Hybrid ★	0.6126	0.3310	0.3112	0.3731	0.5844	0.8017	15.80





Results: MIND Large Development Set

Model	AUC	MRR	nDCG@5	nDCG@10	Div/cat	Div/sub	Novelty
Baseline	0.5385	0.2618	0.2453	0.3079	0.6937	0.8802	14.96
Collaborative	0.5541	0.2795	0.2577	0.3223	0.6681	0.8733	15.64
Content-Based	0.6059	0.3300	0.3097	0.3711	0.5817	0.7977	17.87
Hybrid ★	0.6084	0.3272	0.3066	0.3692	0.6079	0.8202	16.59

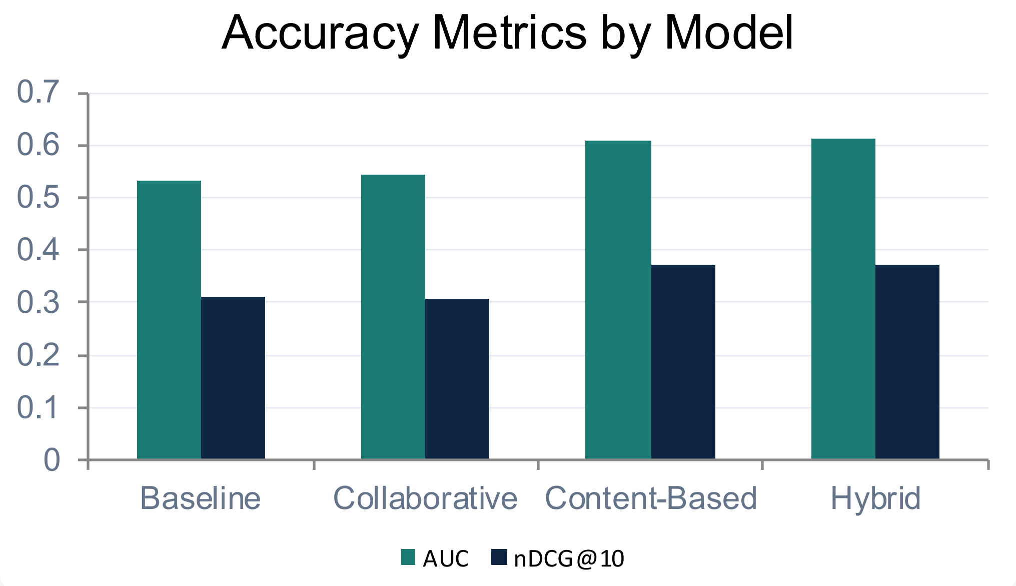
CF improves with more data

On MIND Large, CF AUC rises to 0.5541 vs 0.5429 on MIND Small — more interaction data enables better SVD decomposition and latent factor quality.

CBF stable across scales

Content-based filtering AUC stays ~0.606 on both datasets — confirms item features (not interaction volume) drive performance. Remains strongest individual model.

Accuracy vs Beyond-Accuracy Trade-off



Clear accuracy ordering: Hybrid > Content-Based > Collaborative > Baseline

Beyond-Accuracy Profile

Model	Novelty	Div/cat
Baseline	14.04	0.690
Collaborative	14.58	0.677
Content-Based	**16.03 ↑**	**0.581 ↓**
Hybrid	15.80	0.584

Key Tension

Higher accuracy → lower diversity
Higher novelty → less popular items
Hybrid balances both objectives

Discussion

Baseline	AUC only marginally above 0.5. High diversity but lowest novelty — surfaces popular but predictable articles. Demonstrates value of personalization.
Collaborative (CF)	Modest AUC gain over baseline. Matrix sparsity limits SVD quality. Cold-start prevents scoring of dev-only articles. Improves clearly on MIND Large — confirms need for scale.
Content-Based (CBF)	Strongest individual model. TF-IDF similarity is highly effective on MIND. Handles cold-start items. Highest novelty but narrowest topical diversity — personalization vs. variety trade-off.
Hybrid (CF=0.1 / CBF=0.9)	Best overall AUC. Small CF admixture adds complementary signal. Beyond-accuracy profile stays close to CBF. Tuning heavily favors CBF due to data sparsity.

Conclusion

Clear accuracy ordering: Hybrid > Content-Based > Collaborative > Baseline

TF-IDF signal strength: Textual similarity is a stronger signal than co-consumption under data sparsity

Hybrid benefit: Small but consistent accuracy gain — CF still carries complementary information

Multi-metric evaluation: No single metric fully characterizes quality. Accuracy and beyond-accuracy do not always match

Hybrid fusion insight: Simple weighted approach is effective, most beneficial when one component dominates

Future Work: Stronger CF (e.g., Neural CF) · LLM/transformer embeddings for CBF · Temporal-aware hybrid (recency/freshness)

Thank You

Questions?

github.com/TrippelEM/tdt4215-mind-recsys

Svein · Mehdi · Maryam · Frikk

TDT4215 Recommender Systems · NTNU · 22.04.2026